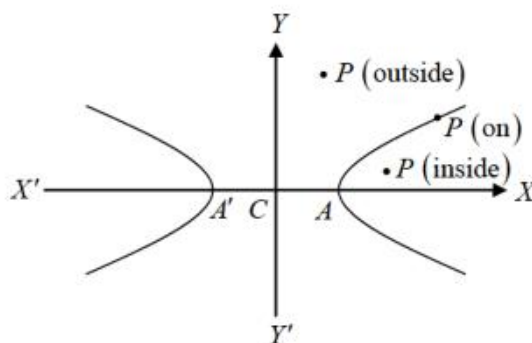


HYPERBOLA

EQUATION	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$	$\frac{(x-a)^2}{a^2} - \frac{(y-b)^2}{b^2} = 1$	$\frac{(y-b)^2}{a^2} - \frac{(x-a)^2}{b^2} = 1$
Centre (C)	(0,0)	(0,0)	(a, b)	(a, b)
Vertices (A, A ¹)	(±a,0)	(0,±a)	(a ± a, b)	(a, b ± a)
Foci (S, S ¹)	(±ae,0)	(0,±ae)	(a ± ae, b)	(a, b ± ae)
Point of intersection of Axis and directrix (Z, Z ¹)	(±a/e,0)	(0,±a/e)	(a ± a/e, b)	(a, b ± a/e)
End point of Latus rectum	(±ae, ±b ² /a)	(±b ² /a, ±ae)	(a ± ae, b ± b ² /a)	(a ± b ² /a, b ± ae)
Equation of Transverse axis	y = 0	x = 0	y = b	x = a
Equation of conjugate axis	x = 0	y = 0	x = a	y = b
Equation of Latera recta	x = ±ae	y = ±ae	x = a ± ae	y = b ± ae
Equation of directrices	x = ±a/e	y = ±a/e	x = a ± a/e	y = b ± a/e
Length of transverse axis	2a	2a	2a	2a
Length of conjugate axis	2b	2b	2b	2b
Length of latus rectum	2b ² /a	2b ² /a	2b ² /a	2b ² /a
Eccentricity (e)	$\sqrt{\frac{a^2 + b^2}{a^2}}$	$\sqrt{\frac{a^2 + b^2}{a^2}}$	$\sqrt{\frac{a^2 + b^2}{a^2}}$	$\sqrt{\frac{a^2 + b^2}{a^2}}$
Difference of focal distances of a point P on the ellipse	2a	2a	2a	2a
Distance between foci	2ae	2ae	2ae	2ae
Distance between Vertices	2a	2a	2a	2a
Distance between Directrix	2a/e	2a/e	2a/e	2a/e

- (1) Equation of conjugate hyperbola of a given hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$.
 - (2) The hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ are called as conjugate hyperbolas to each other.
 - (3) If e_1 and e_2 are the eccentricity of two conjugate hyperbolas then $e_1^2 + e_2^2 = e_1^2 e_2^2$ or $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$.
 - (4) The focus of a hyperbola and conjugate hyperbola are concyclic. i.e. $ae = be'$ where e is eccentricity of hyperbola and e' is eccentricity of conjugate hyperbola.
- **Rectangular Hyperbola:** A hyperbola is said to be a rectangular hyperbola if the length of its transverse axis is equal to the length of its conjugate axis.
 - (1) $x^2 - y^2 = a^2$ and $xy = c^2$ are the rectangular hyperbolas.
 - (2) The eccentricity of rectangular hyperbola is $\sqrt{2}$.
 - (3) The parametric form of $xy = c^2$ are $x = ct$, $y = \frac{c}{t}$.
 - **Position of a point with respect to a hyperbola:** Let $P(x_1, y_1)$ be a point and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be an hyperbola, then
 - (a) $P(x_1, y_1)$ lies on hyperbola if $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1 = 0$.
 - (b) $P(x_1, y_1)$ lies inside hyperbola if $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1 > 0$.
 - (c) $P(x_1, y_1)$ lies outside hyperbola if $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1 < 0$.



- **Intersection of a line and a hyperbola:** The straight line $y = mx + c$ will cut the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ in two points:
 - (1) May be real, if $c^2 > a^2 m^2 - b^2$.
 - (2) May be coincident, if $c^2 = a^2 m^2 - b^2$. (condition of Tangency)

(3) May be imaginary, if $c^2 < a^2 m^2 - b^2$.

• **Equation of Tangent in Different Forms:**

(1) From an external point we can draw only two tangents to a hyperbola.

(2) **Point Form:** The equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $P(x_1, y_1)$ is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1 = 0$.

(3) **Parametric Form:** The equation of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $(a \sec \theta, b \tan \theta)$ is $\frac{x}{a} \sec \theta + \frac{y}{b} \tan \theta = 1$.

(4) **Slope Form:** The equation of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ may be taken as $y = mx \pm \sqrt{a^2 m^2 - b^2}$ where m is the slope of the tangent. Point of contact is $\left(\pm \frac{a^2 m}{\sqrt{a^2 m^2 - b^2}}, \pm \frac{b^2}{\sqrt{a^2 m^2 - b^2}} \right)$.

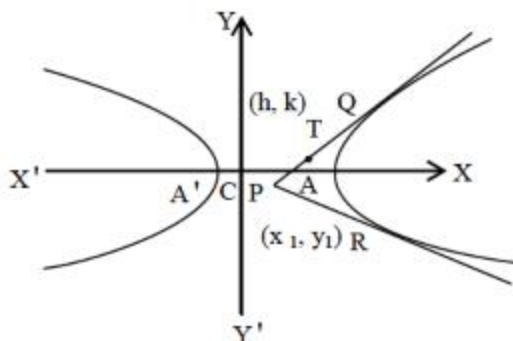
(5) The condition of the line $y = mx + c$ may be a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $c^2 = a^2 m^2 - b^2$ and the point of contact is $\left(\frac{-a^2 m}{c}, \frac{-b^2}{c} \right)$.

(6) The condition of the line $lx + my + n = 0$ may be a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $a^2 l^2 - b^2 m^2 = n^2$ and the point of contact is $\left(-\frac{a^2 l}{n}, \frac{b^2 m}{n} \right)$.

(7) If m_1 and m_2 are the slopes of the tangents from a point $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then $m_1 + m_2 = \frac{2x_1 y_1}{x_1^2 - a^2}$ and $m_1 m_2 = \frac{y_1^2 + b^2}{x_1^2 - a^2}$.

(8) If θ is the angle between the tangents from $P(x_1, y_1)$ to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then
$$\tan \theta = \frac{2ab \sqrt{1 - \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}}}{x_1^2 + y_1^2 - a^2 + b^2}.$$

(9) The equation of pair of tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is
$$\left(\frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1 \right)^2 = \left(\frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 \right) \left(\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1 \right).$$



• **Equation of Normal in Different Forms:**

- (1) **Point Form:** The equation of the normal at $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2.$$

- (2) Equation of normal at the point $P(x_1, y_1)$ to the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $\frac{x - x_1}{ax_1 + hy_1 + g} = \frac{y - y_1}{hx_1 + by_1 + f}.$

- (3) **Parametric Form:** The equation of normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $(a \sec \theta, b \tan \theta)$ is $ax \cos \theta + by \cot \theta = a^2 + b^2.$

- (4) **Slope Form:** The equation of normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ in terms of slope m of normal is $y = mx \pm \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2 m^2}}.$

- (5) **Condition of Normality:** The condition of the line $y = mx + c$ may be a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $c^2 = \frac{m^2(a^2 + b^2)^2}{a^2 - m^2 b^2}.$

- (6) The condition that the line $lx + my + n = 0$ to be normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}.$

- (7) At most Four normals can be drawn from any point to the hyperbola.

• **Director Circle:** The locus of the point of intersection of the perpendicular tangents of a hyperbola is a circle concentric with the hyperbola. It is known as Director Circle.

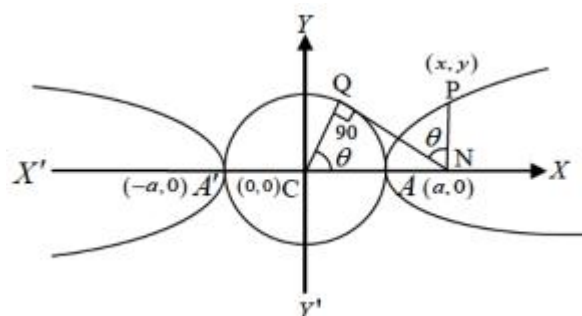
- (1) The equation of the director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 - b^2.$

- (2) For the hyperbola $\frac{(x-a)^2}{a^2} - \frac{(y-b)^2}{b^2} = 1$, equation of the director circle is $(x-a)^2 + (y-b)^2 = a^2 - b^2$.

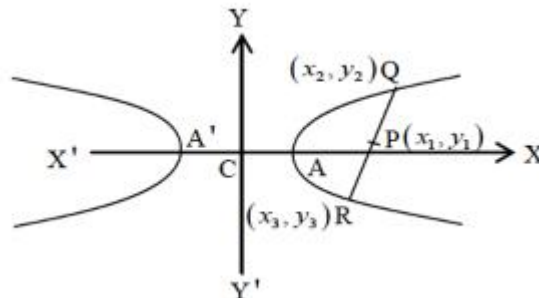
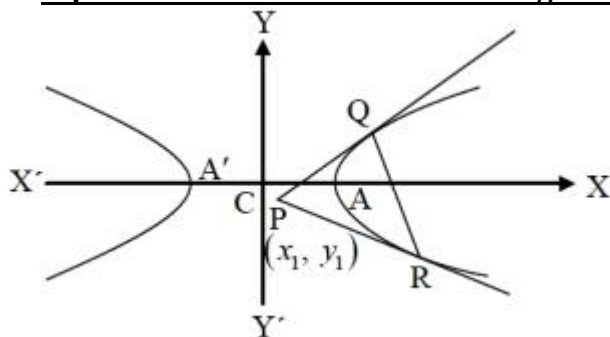
- **Auxillary Circle:** The locus of the feet of the perpendicular drawn from the foci to any tangent of the hyperbola is a circle concentric with the hyperbola. It is known as Auxillary Circle.

- (1) Let $\angle QCN = \theta$. Here P and Q are the corresponding points on the hyperbola and the auxillary circle $0 \leq \theta \leq 2\pi$.

- (2) The equation of the Auxillary circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2$.



- **Equation of Chord of Contact of tangents drawn from a point to the hyperbola:**



- (1) Let PQ and PR be tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ drawn from any external point $P(x_1, y_1)$, then the equation of chord of contact QR is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1 = 0$.

- (2) The equation of chord of hyperbola $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ having (x_1, y_1) as its midpoint is $\left(\frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1\right) = \left(\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1\right)$.

- (3) The equation of chord joining the points a and b on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{x}{a} \cos \frac{a-b}{2} - \frac{y}{b} \sin \frac{a+b}{2} = \cos \frac{a+b}{2}$.

- (4) If a and b are the ends of focal chord of an hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then

$$e \cos \frac{a-b}{2} = \cos \frac{a+b}{2}.$$

- (5) The tangents at the ends of the focal chord meet on the directrix.

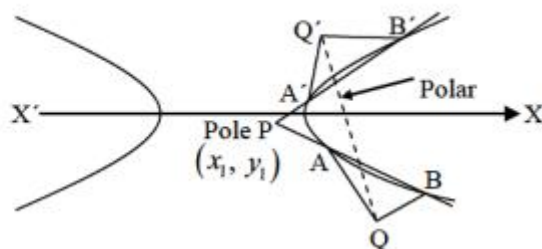
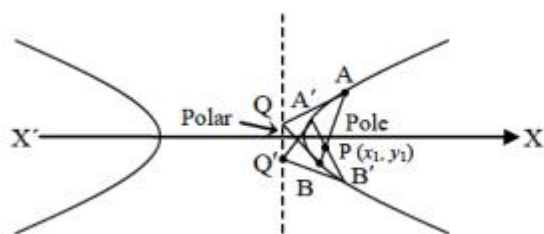
- (6) Points of intersection of tangents at $P(a \sec q_1, b \tan q_1)$ and $Q(a \sec q_2, b \tan q_2)$ is

$$\left(a \frac{\cos\left(\frac{q_1 - q_2}{2}\right)}{\cos\left(\frac{q_1 + q_2}{2}\right)}, b \frac{\sin\left(\frac{q_1 + q_2}{2}\right)}{\cos\left(\frac{q_1 + q_2}{2}\right)} \right).$$

- (7) Points of intersection of normals at $P(a \sec q_1, b \tan q_1)$ and $Q(a \sec q_2, b \tan q_2)$ is

$$\left(\frac{a^2 + b^2}{a} \sec q_1 \sec q_2 \frac{\cos\left(\frac{q_1 - q_2}{2}\right)}{\cos\left(\frac{q_1 + q_2}{2}\right)}, -\frac{a^2 + b^2}{a} \tan q_1 \tan q_2 \frac{\sin\left(\frac{q_1 + q_2}{2}\right)}{\cos\left(\frac{q_1 + q_2}{2}\right)} \right).$$

- **Pole and Polar of Hyperbola:** The locus of the point of intersection of tangents at the ends of a variable chord drawn from a fixed point P on the hyperbola is called as the polar of the given point P with respect to the hyperbola and the point P is called as Pole of the polar.



- (1) The equation of polar of the point $P(x_1, y_1)$ with respect to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1 = 0.$$

- (2) The pole of the line $lx + my + n = 0$ with respect to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

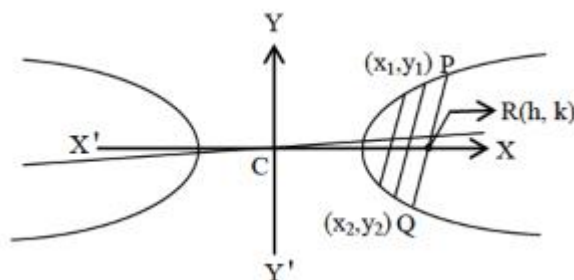
$$\left(\frac{-a^2 l}{n}, \frac{b^2 m}{n} \right), n \neq 0.$$

- (3) The pole of the line $lx + my + n = 0$ with respect to the hyperbola

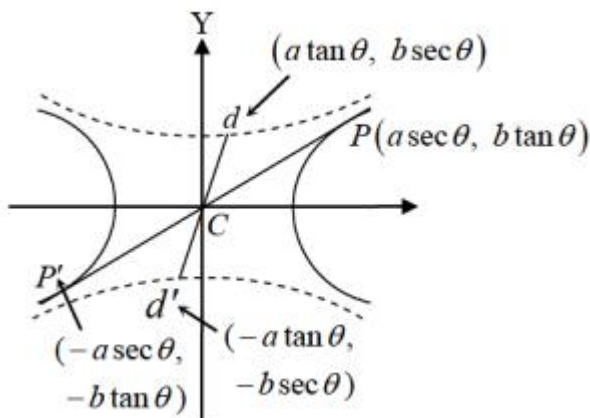
$$\frac{(x-a)^2}{a^2} - \frac{(y-b)^2}{b^2} = 1 \text{ is } \left(a - \frac{a^2 l}{la + mb + n}, b + \frac{b^2 m}{la + mb + n} \right).$$

- (4) If the polar $P(x_1, y_1)$ passes through $Q(x_2, y_2)$, then the polar of $Q(x_2, y_2)$ passes through $P(x_1, y_1)$ and such points are called as **conjugate points**.

- (5) If the pole of a line $l_1x + m_1y + n_1 = 0$ lies on the another line $l_2x + m_2y + n_2 = 0$, then the pole of line $l_2x + m_2y + n_2 = 0$ lies on line $l_1x + m_1y + n_1 = 0$. Such lines are called as **conjugate lines**.
 - (6) Pole of a given line is same as point of intersection of tangents as its extremities.
 - (7) The condition for two points (x_1, y_1) and (x_2, y_2) to be conjugate with respect to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{x_1x_2}{a^2} - \frac{y_1y_2}{b^2} - 1 = 0$.
 - (8) The condition that two lines $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ to be conjugate with respect to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $a^2l_1l_2 - b^2m_1m_2 = n_1n_2$.
 - (9) If H, C and A are the equations of a hyperbola, its conjugate hyperbola and its pair of asymptotes respectively then $H + C = 2A$.
- **Diameter of Hyperbola:** The locus of the middle point of a system of parallel chords of a hyperbola is called as diameter and the point where the diameter intersects the hyperbola is called as vertex of the hyperbola.

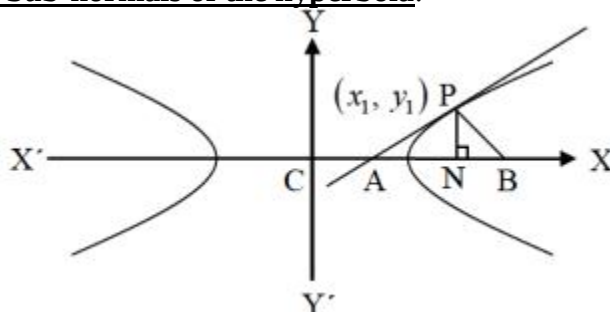


- (1) Let $y = mx + c$ may be system of parallel chords to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for different chords then the equation of diameter of the hyperbola is $y = \frac{b^2x}{a^2m}$ which passes through the origin (0, 0).
- **Conjugate Diameter:** Two diameters are said to be conjugate diameter when each bisects all chords parallel to the other.



- (1) If $y = m_1x$ and $y = m_2x$ be conjugate diameters, then $m_1m_2 = \frac{b^2}{a^2}$.
- (2) If the two extremities of a diameter lie in the first and third quadrants, the extremities of the conjugate diameters also lie in the first and third quadrants.
- (3) The extremities d and d' of the conjugate diameter do not lie on the hyperbola.

• **Sub-tangents and Sub-normals of the hyperbola:**

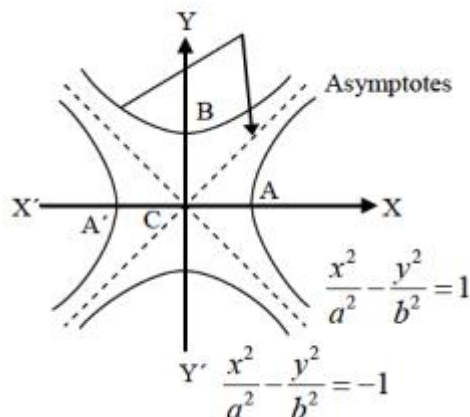


Let the tangent and normal at point $P(x_1, y_1)$ meets the x-axis at A and B respectively.

- (1) Length of Sub-tangent $AN = CN - CA = x_1 - \frac{a^2}{x_1}$.
 - (2) Length of Sub-normal $BN = CB - CN = \frac{a^2 + b^2}{a^2} x_1 - x_1 = \frac{b^2}{a^2} x_1 = (e^2 - 1)x_1$.
- **Asymptote to the Hyperbola:** If the length of the perpendicular let fall from a point on the hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called an asymptote of the hyperbola.
- **Properties of Asymptotes of Hyperbola:**

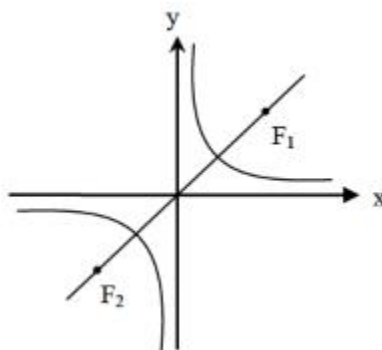
- (1) The equation of asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $\frac{x}{a} \pm \frac{y}{b} = 0$ or $y = \pm \frac{b}{a}x$.
- (2) The equation of pair of asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.
- (3) Equation of hyperbola and equation of its pair of asymptotes will differ in their constant terms only.
- (4) Asymptote of the hyperbola passes through the centre of the hyperbola and they are equally inclined to the axes of the hyperbola.
- (5) The angle of the asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $2 \tan^{-1}\left(\frac{b}{a}\right)$ or $2 \sec^{-1} e$.
- (6) The angle between the asymptotes of rectangular hyperbola is $\frac{\pi}{2}$.

- (7) The product of perpendiculars from any point on hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to its asymptotes is $\frac{a^2 b^2}{a^2 + b^2}$.



- (8) The asymptotes of a hyperbola and its conjugate hyperbola are same.
 (9) When $a = b$ i.e. the asymptote of the rectangular hyperbola $x^2 - y^2 = a^2$ are $y = \pm x$ which are at right angles.
 (10) The asymptotes pass through the centre of the hyperbola.
 (11) The bisector of the angle between the asymptotes is the co-ordinate axes.
 (12) Asymptotes are equally inclined to the axes of the hyperbola.

• **For rectangular hyperbola or equilateral hyperbola $xy = c^2$:**



- (a) A hyperbola whose asymptotes are at right angles to each other is called as a rectangular hyperbola.
 (b) The general equation of second degree represents a rectangular hyperbola if $\Delta \neq 0$, $h^2 > ab$ and coefficient of $x^2 + \text{coefficient of } y^2 = 0$.
 (c) Asymptotes are $x = 0$, $y = 0$.
 (d) Transverse axis $y = x$.
 (e) Conjugate axis $y = -x$.
 (f) Vertex $A(c, c)$ and $A'(-c, -c)$.
 (g) Foci $(c\sqrt{2}, c\sqrt{2})$ and $(-c\sqrt{2}, -c\sqrt{2})$.
 (h) Length of latus rectum $= 2\sqrt{2}c$.

- (i) Equation of auxillary circle: $x^2 + y^2 = 2c^2$.
- (j) Equation of director circle is $x^2 + y^2 = 0$.
- (k) $x^2 - y^2 = 1$ and $xy = c^2$ intersect at right angle.
- (l) Eccentricity of rectangular hyperbola is $\sqrt{2}$.
- (m) Parametric form of rectangular hyperbola is $x = ct, y = \frac{c}{t}$ where $t \in R - \{0\}$.
- (n) Slope of chord joining the point $\left(ct_1, \frac{c}{t_1}\right)$ and $\left(ct_2, \frac{c}{t_2}\right)$ is $\frac{-1}{t_1 t_2}$.
- (o) Slope of tangent at $\left(ct, \frac{c}{t}\right)$ is $\frac{-1}{t^2}$.
- (p) Equation of tangent at point whose parameter 't' is $y - \frac{c}{t} = \frac{-1}{t^2}(x - ct)$ or $x + yt^2 - 2ct = 0$.
- (q) Equation of normal at point whose parameter 't' is $y - \frac{c}{t} = t^2(x - ct)$ or $xt^3 - yt - ct^4 + c = 0$.
- (r) Equation of tangent at (x_1, y_1) is $xy_1 + yx_1 = 2c^2$.
- (s) Equation of normal at (x_1, y_1) is $xy_1 - yx_1 = x_1^2 - y_1^2$.
- (t) Chord with given middle point as (x_1, y_1) is $xy_1 + yx_1 = 2x_1y_1$.
- (u) If $\left(ct_1, \frac{c}{t_1}\right), \left(ct_2, \frac{c}{t_2}\right)$ and $\left(ct_3, \frac{c}{t_3}\right)$ are point on $xy = c^2$ then the orthocenter of the triangle formed by them is $\left(\frac{-c}{t_1 t_2 t_3}, -ct_1 t_2 t_3\right)$.
- (v) If a circle and the rectangular hyperbola $xy = c^2$ meet at the four points t_1, t_2, t_3, t_4 then $t_1 t_2 t_3 t_4 = 1$.
- (w) The centre of the mean position of the four points bisects the distance between the centres of the two curves.

• **Equation of tangents of rectangular hyperbola in different form:**

- (1) **Point Form:** The equation of tangent to the rectangular hyperbola $xy = c^2$ at $P(x_1, y_1)$ is $xy_1 + yx_1 = 2c^2$ or $\frac{x}{x_1} + \frac{y}{y_1} = 2$.
- (2) **Parametric Form:** The equation of tangent to the rectangular hyperbola $xy = c^2$ at $P\left(ct, \frac{c}{t}\right)$ is $\frac{x}{t} + yt = 2c \Rightarrow x + yt^2 = 2ct$.

- (3) The point of intersection of tangents at $\left(ct_1, \frac{c}{t_1}\right)$ and $\left(ct_2, \frac{c}{t_2}\right)$ is $\left(\frac{2ct_1t_2}{t_1+t_2}, \frac{2c}{t_1+t_2}\right)$.
- **Equation of Normal in different Form:**
 - (1) **Point Form:** The equation of normal to the rectangular hyperbola $xy = c^2$ at $P(x_1, y_1)$ is $xy_1 - yx_1 = x_1^2 - y_1^2$.
 - (2) **Parametric Form:** The equation of normal to the rectangular hyperbola $xy = c^2$ at $P\left(ct, \frac{c}{t}\right)$ is $xt^3 - yt - ct^4 + c = 0$.
 - (3) The point of intersection of normals at $\left(ct_1, \frac{c}{t_1}\right)$ and $\left(ct_2, \frac{c}{t_2}\right)$ is $\left(\frac{c\{t_1t_2(t_1^2 + t_1t_2 + t_2^2) + 1\}}{t_1t_2(t_1 + t_2)}, \frac{c\{t_1^3t_2^3 + (t_1^2 + t_1t_2 + t_2^2)\}}{t_1t_2(t_1 + t_2)}\right)$.
 - (4) We can draw four normals from a point to the rectangular hyperbola $xy = c^2$.
 - **Diameter of Rectangular Hyperbola:** Equation of diameter of rectangular hyperbola $xy = c^2$ is $y + mx = 0$ where m is the slope of the chord joining two points' lies on the rectangular hyperbola.
 - **Conjugate diameters of rectangular hyperbola:** Two diameters $y + m_1x = 0$ and $y + m_2x = 0$ are conjugate diameter if $m_1 + m_2 = 0$.
 - **Properties of Hyperbola** $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right)$:
 - (1) If PN be the ordinate of a point P on the hyperbola and the tangent at P meets the transverse axis in T, then $ON \cdot OT = a^2$, O is the origin.
 - (2) If PM be drawn perpendiculars to the conjugate axis from a point P on the hyperbola and the tangent at P meets the conjugate axis in T, then $OM \cdot OT = -b^2$, O is the origin.
 - (3) If the normal at P on the hyperbola meets the transverse axis in G, then $SG = eSP$; S is the foci and e is the eccentricity of the hyperbola.
 - (4) The tangent and normal at any point of hyperbola bisect the angle between the focal radii to that point.
 - (5) The locus of the feet of the perpendicular from the foci on a tangent to a hyperbola is the auxiliary circle.
 - (6) The product of the length of the perpendicular drawn from foci on any tangent to the hyperbola is b^2 .

- (7) From any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ perpendicular are drawn to the asymptotes then product is $\frac{a^2 b^2}{a^2 + b^2}$ and for the rectangular hyperbola $\frac{a^2}{2}$.
- (8) If a circle cuts the rectangular hyperbola $xy = 1$ in (x_r, y_r) four points $r = 1, 2, 3, 4$ then $\frac{1}{x_1 x_2 x_3 x_4} = y_1 y_2 y_3 y_4 = 1$.
- (9) A rectangular hyperbola with centre at C is cut by any circle of radius R in four points L, M, N, P then the value of $CL^2 + CM^2 + CN^2 + CP^2 = 4R^2$.
- (10) If a triangle is inscribed in a rectangular hyperbola then the orthocenter of triangle lies on the rectangular hyperbola.
- (11) The portion of tangent intercepted between the asymptotes at any point of the hyperbola is bisected by the point of contact.
- (12) Whenever any circle and any hyperbola cut each other at four points the mean position of these four points is the midpoint of the line segment joining centre of hyperbola and centre of circle.
- (13) The harmonic mean of focal radii for any focal chord = $\frac{b^2}{2}$.
- (14) Tangent drawn at the ends of any focal chord meet on the directrix.
- (15) Locus of point of intersection two perpendicular tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a circle is called as director circle whose equation is $x^2 + y^2 = a^2 - b^2$ and if $a < b$ then there is no real point from where we can draw two perpendicular tangents to the hyperbola.
- (16) The portion of tangent between point of contact and the point where it cuts the directrix subtend 90° angle at the focus.
